Treat product codes as information to verify

A product code can identify a grade system or a product, but it does not supply the whole application instruction. A penetration-grade designation and a performance-grade designation are different ways of describing binder characteristics. The numbers are not interchangeable merely because both appear beside asphalt. The purchaser and operator need the approved product, governing specification, supplier record and relevant safety information.

Fictional document-check case. An archive card contains the labels “Group A / C,” “85–100 Pen” and “PG 72–22,” and a handwritten note says “altered oil.” A new practice request says only “send the usual oil.” The meaning of “Group A / C” is not defined in this archive card. No current approved binder, project specification or supplier product sheet has been attached. These labels are for recognition and clarification; the case does not recommend either grade for B.C. work. Do not infer a mixing temperature, quantity or handling procedure from them.

Write the clarification before the order

First identify which archive wording is undefined and must be clarified; do not guess a classification from the letters. Make a four-item information request that would identify the intended product and its governing documents. Explain why “usual oil” and an archive grade label are insufficient. Read your request to a partner, who checks whether you have accidentally approved a product. Keep it as a classroom draft. **Skills:** Writing, Communication, Problem Solving.

Separate the recycling routes

Reclaimed asphalt pavement (RAP) is asphalt pavement material recovered for possible reuse. A plant-recycling route takes recovered material through suitable processing and an approved production process. Reuse is conditional on the design and quality requirements. A claim that all recovered material can be added without testing is unsupported.

Hot-in-place recycling treats an existing asphalt surface on the road using a coordinated equipment train. Its broad purpose is to reuse existing material in place through a specialist process. It is different from hauling recovered material to a plant. A milling machine's removal of material is also not proof that the removed material has been successfully recycled. Track the actual route and evidence for reuse.

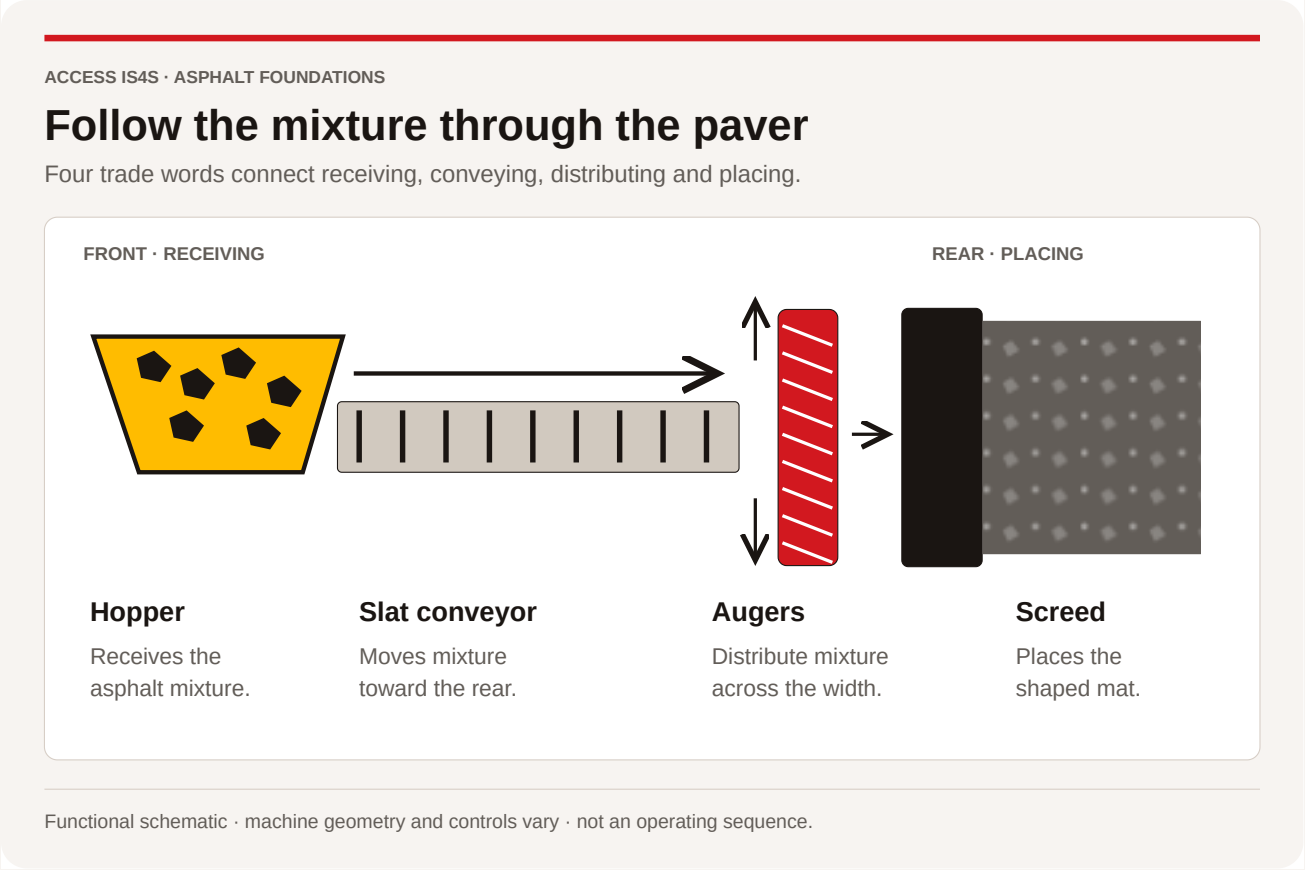
An **asphalt train** is a coordinated sequence of equipment and roles. In ordinary placement, haul, paver/screed and compaction functions connect. A specialist hot-in-place train has its own process and equipment. The word *train* does not mean every paving job uses the same machinery, controls or sequence.

Follow the mixture inside the paver

The **hopper** receives mixture. **Slat conveyors** move it toward the rear. **Augers** distribute it across the width ahead of the **screed**, which places and shapes the mat. The screed also performs the placement stage's initial compaction; rollers have a later compaction role. Read each noun together with its action. The material route explains a general function; an identified machine manual supplies the actual layout and controls.

Figure LA-5

Follow the mixture through the paver



Note. Adapted from *The asphalt handbook* (7th ed., MS-4; Figure 8.11, p. 377 / PDF 411; 8.4.1, component descriptions, p. 378 / PDF 412), by Asphalt Institute, 2007. Replaced the perspective equipment schematic with a labelled functional material route. Not drawn to scale or tied to an actual model. Original Figure 8.11 is credited to Volvo Construction Equipment in the handbook's contributor list. Copyright 2007 by Asphalt Institute; original contributor credits retained.

Worked explanation. "The conveyor moves mixture rearward; the auger distributes it across the width." The direction and job differ. Find the hopper and screed in the manufacturer photograph next, then explain why a general route diagram cannot establish another machine's numbered callouts.

Read one machine picture carefully

The flow map names a function. A manufacturer picture helps you locate a particular machine's parts. Start with the model and numbered callout, then read the matching name. This example introduces paver component vocabulary for later equipment study.

Figure LA-3

Find the named part on a LeeBoy 8680 paver

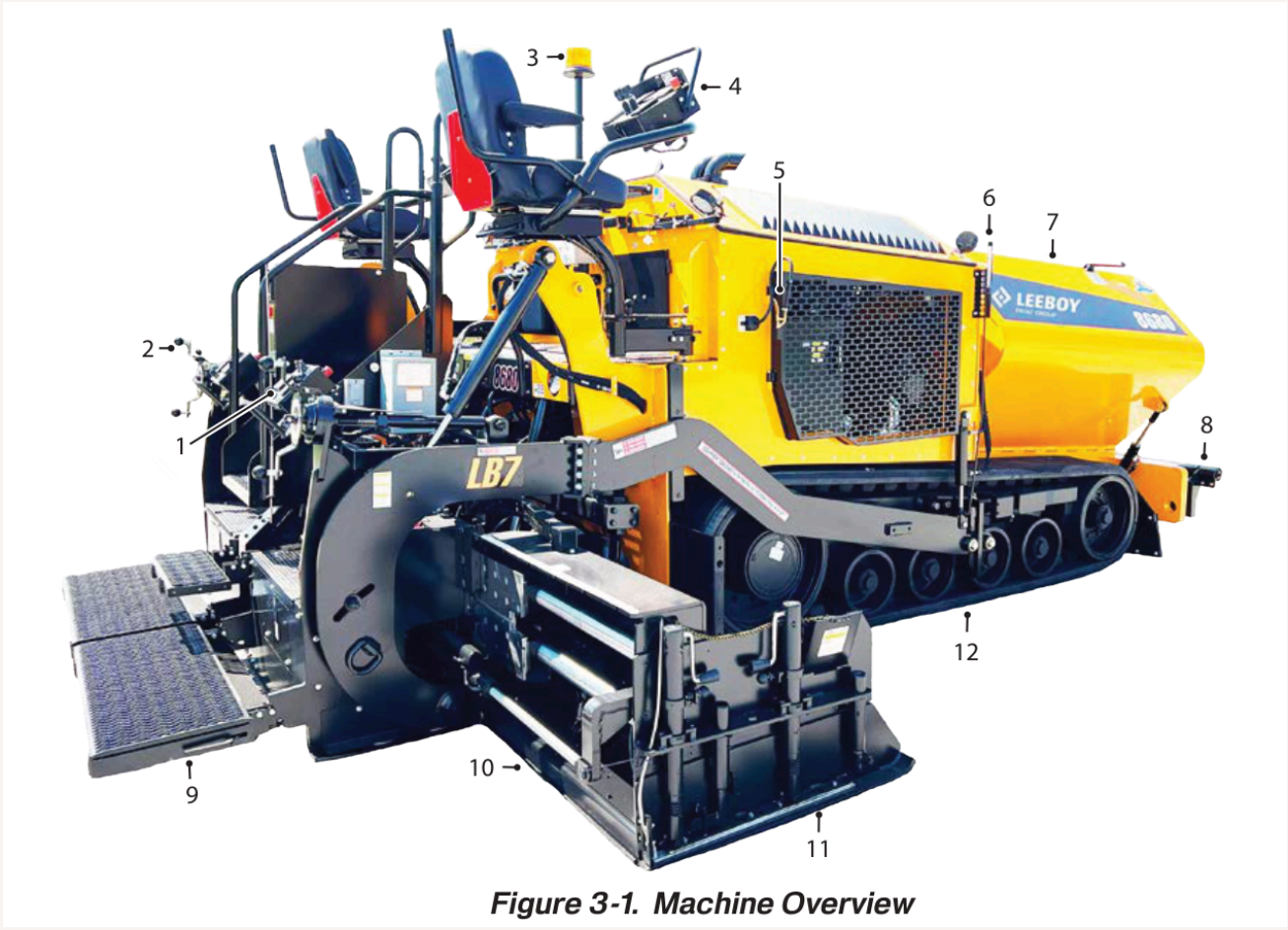


Figure 3-1. Machine Overview

Model 8680 · Manual 1033428-SOP · Serial number 365152 and above

Note. Cropped from Figure 3-1, "Machine Overview," in *LeeBoy model 8680 commercial paver: Safety and operations manual* (Manual No. 1033428-SOP, p. 3-7; PDF p. 43), by LeeBoy, 2026 (manufacturer publication). Copyright 2026 by LeeBoy. Original numbered callouts retained; the surrounding page and legend are omitted from the crop. Selected original component names appear below; the plain-language explanations are authored for this lesson. Open the complete source page and legend.

Original callout	Manufacturer's name	Plain-language explanation
7	Hopper	Holds the supplied asphalt material on this paver
10	Screed Extension	Extends the screed's working width
11	End Gate	Contributes to forming the edge of the placed mat
12	High-Speed Rubber Tracks	The rubber tracks on which this machine travels; the source name supplies no permission to select a travel speed

Worked example · Look → name → explain. "Callout 7 points to the **hopper**. It holds the supplied asphalt material. The photo does not tell me the approved mixture, the operating setting or whether another paver has the same arrangement." Use the selected table to explain a second part. Keep the manual identity with your answer. The complete manual has the other callouts and their context.

Trace two reuse pathways

Draw two labelled routes: recovered material sent to a plant, and a specialist hot-in-place process on the existing road. Mark where the existing material is recovered or reused. Add one document or check needed before calling either route approved. Explain why “milled” and “recycled into a new pavement” are different claims. **Skills:** Reading, Creativity and Innovation, Problem Solving.

Calculate a fictional material record

A classroom demonstration model uses **80 counters for new aggregate, 15 for recovered mixture and 5 for binder**, for 100 counters in total. Counters are a visual model, not a mix recipe; the recovered mixture already contains its own materials. Find each category's share of the model and check the sum. Explain why adding the five binder counters to the 15 recovered-mixture counters does **not** establish the real binder content. **Skills:** Numeracy, Reading.

Respond to a changed process description

A team initially labels a photo “RAP going to a plant.” The caption is corrected to “specialist hot-in-place train.” Revise the material-flow card and give a handover identifying what changed, what remains accurate and what further process information would be needed. Each learner makes one recorded revision. **Skills:** Adaptability, Collaboration, Communication.

Make a retrievable technology note

Create a one-page document with the two plant distinctions, ingredient roles, two recycling routes and the unresolved product request. Save, close and reopen it; check that another learner can find its title and version. Offer a paper route for composing, while keeping actual Digital evidence separate. Finish by explaining one function without providing an operating setting. Add a small Figure LA-3 note: choose two of callouts 7, 10, 11 or 12, record the manufacturer’s names, explain each in plain language, and retain the model, manual number and page. State one thing the photograph cannot establish. Add the four-stage material route from Figure LA-5, using the words hopper, slat conveyors, augers and screed. Explain how the conveyor and auger functions differ. Use Figure LA-4 to check that your plant comparison describes the mixing process.

Use AI to create and audit the note. ChatGPT or Claude can help explain the supplied text, draft a document and review a draft. Claude Code or Codex can help build an HTML page or SVG; Cowork can help with approved files and folders when that capability is available. Choose the tool for the task and check its current file capabilities.

1. Give the assistant only the fictional case and approved source extracts. State the audience, required facts, missing information and requested files. Ask it to flag gaps instead of inventing settings.
2. Request an editable DOCX note or Markdown draft, a labelled SVG of the material route, and a PDF or HTML reading copy. Keep the source title, edition and page with every technical claim or figure. Renaming a file extension does not convert its content.
3. Ask for an audit table: **claim → source/page → supported, incorrect or unresolved → correction**. Check the table yourself against the actual documents. Another AI response is not independent technical approval.

4. Make a 45–60 second study script with a question, pause and checked answer. Use a suitable audio tool for MP3/WAV, or a suitable video tool for MP4/WebM. Keep a TXT transcript and SRT/VTT captions. A script or storyboard alone is not a generated recording.
5. Open the actual files, inspect every label and play the whole recording. Correct invented parts, unreadable captions or missing qualifications. Save the checked version and have a partner explain the material route from it.

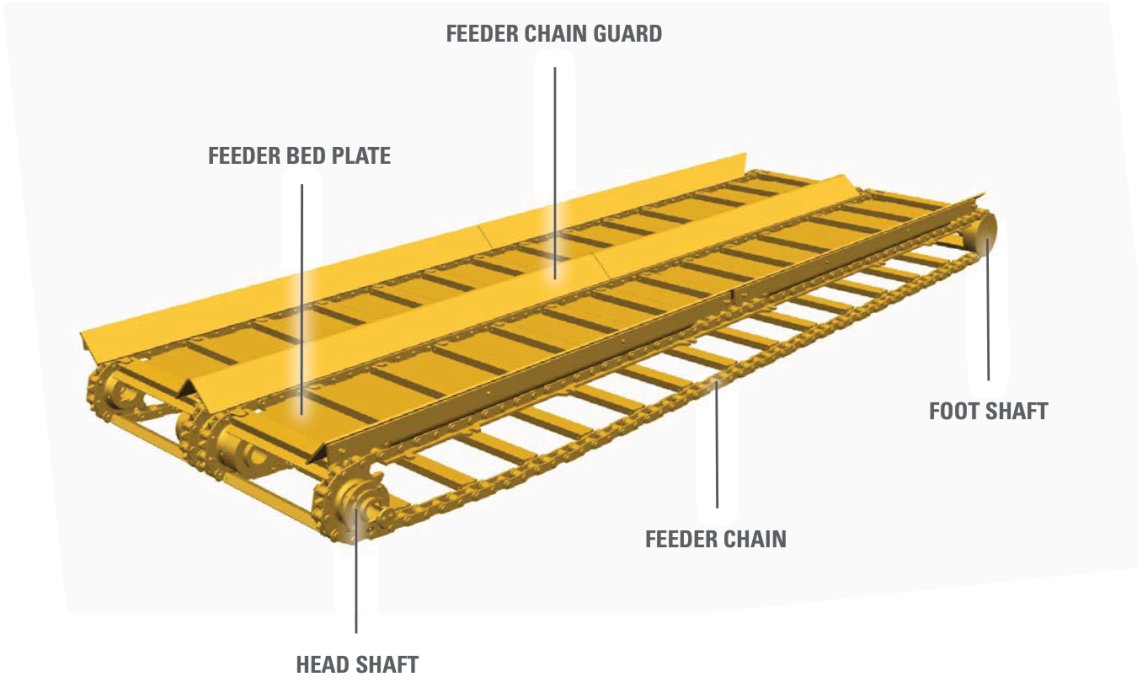
For a spreadsheet calculation, request XLSX formulas and a CSV data copy; check units and totals after reopening. For slides or revision cards, PPTX and CSV/TSV may be useful. Pick the format for the next person's actual task, preserve the editable master and label the reviewed reading copy.

Inspect the route at component level

The **feeder system** moves mixture through the paver. A feeder chain, its shafts and the bed plate have different functions: a chain transmits movement, shafts support or drive the assembly as designed, and the bed plate forms a wear surface. The **auger assembly** moves mixture across the width near the screed. The terms below belong to the manufacturer's illustrated parts families; actual configurations must be matched to the machine.

FIGURE T08

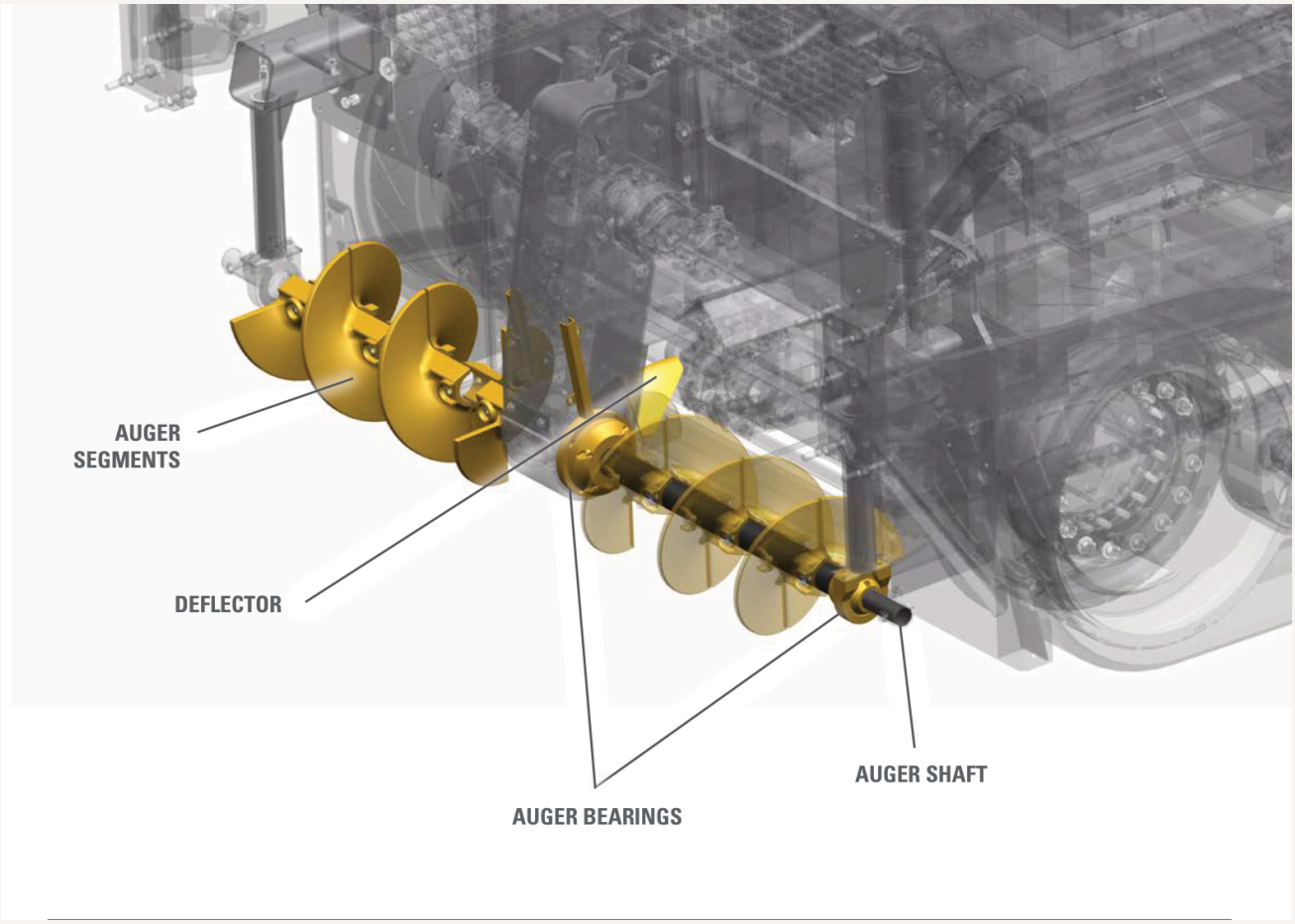
Inside the paver: the feeder system



Note. From Pavers and Screeds Parts Reference Guide (QEXQ3132-02 (08/2024), source PDF p. 11), by Caterpillar, 2024. Copyright Caterpillar. Faithful 200 dpi colour crop; original diagrams, callouts and labels retained. No labels invented or replaced. Complete source-page link accompanies the figure. Scope: Multiple paver/screed models and build numbers; applicability is specified in individual tables. Source and context.

FIGURE T09

Auger parts: identify the assembly



Note. From Pavers and Screeds Parts Reference Guide (QEXQ3132-02 (08/2024), source PDF p. 15), by Caterpillar, 2024. Copyright Caterpillar. Faithful 200 dpi colour crop; original diagrams, callouts and labels retained. No labels invented or replaced. Complete source-page link accompanies the figure. Scope: Multiple paver/screed models and build numbers; applicability is specified in individual tables. Source and context.

Trace movement and identify parts

On the two CAT illustrations identify feeder bed plate, feeder chain, foot shaft, head shaft, auger segments and auger shaft. Explain which picture concerns rearward feeding and which concerns lateral distribution. Write one sentence distinguishing a wear surface from a moving assembly. This is a picture-identification exercise; inspection or maintenance around these mechanisms requires the actual isolation procedure and qualified instruction. **Skills:** Reading, Writing, Communication.

Read the screed as a responding system

A free-floating screed is pulled by the tractor and responds to interacting forces and material conditions. The handbook drawing names forward motion, screed weight, head of material, angle of attack and material stability. **Head of material** describes the material in front of the screed. **Angle of attack** describes the screed's angle in this system. A change in one condition can affect the others and the resulting mat. A learner should be able to name these relationships before interpreting an adjustment procedure.

FIGURE T10

Read the forces around a free-floating screed

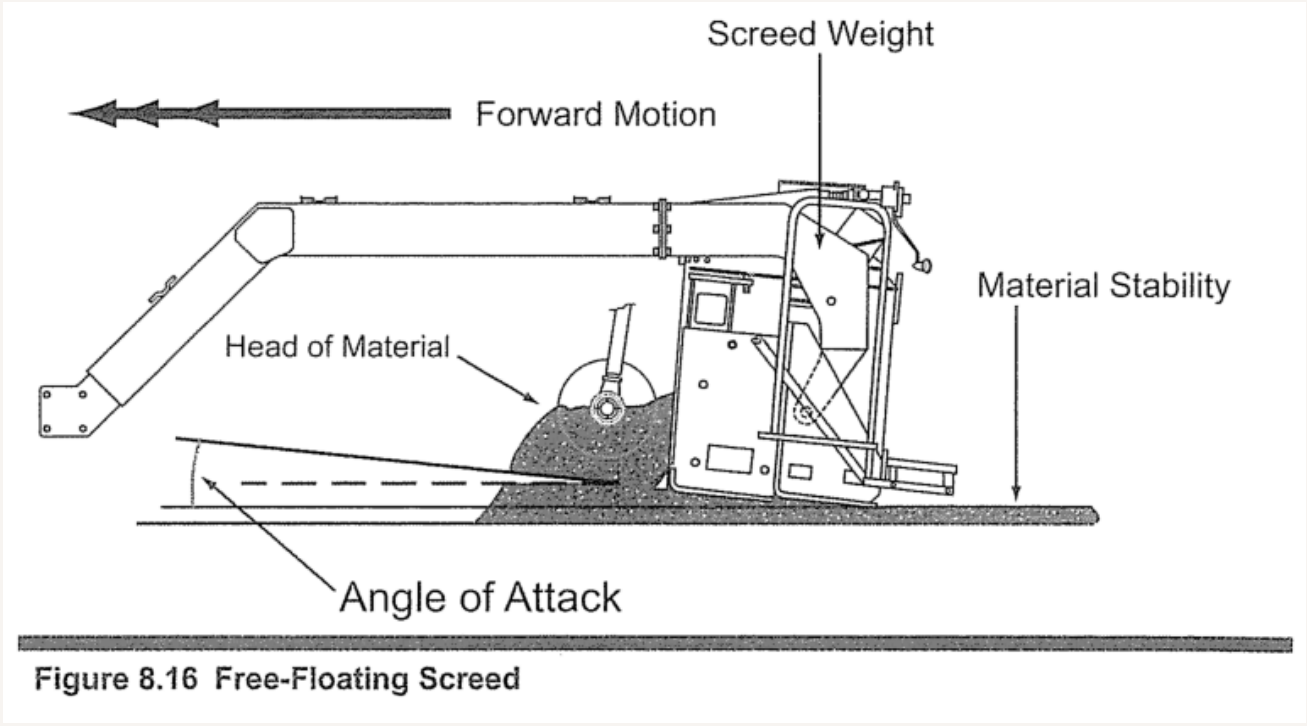
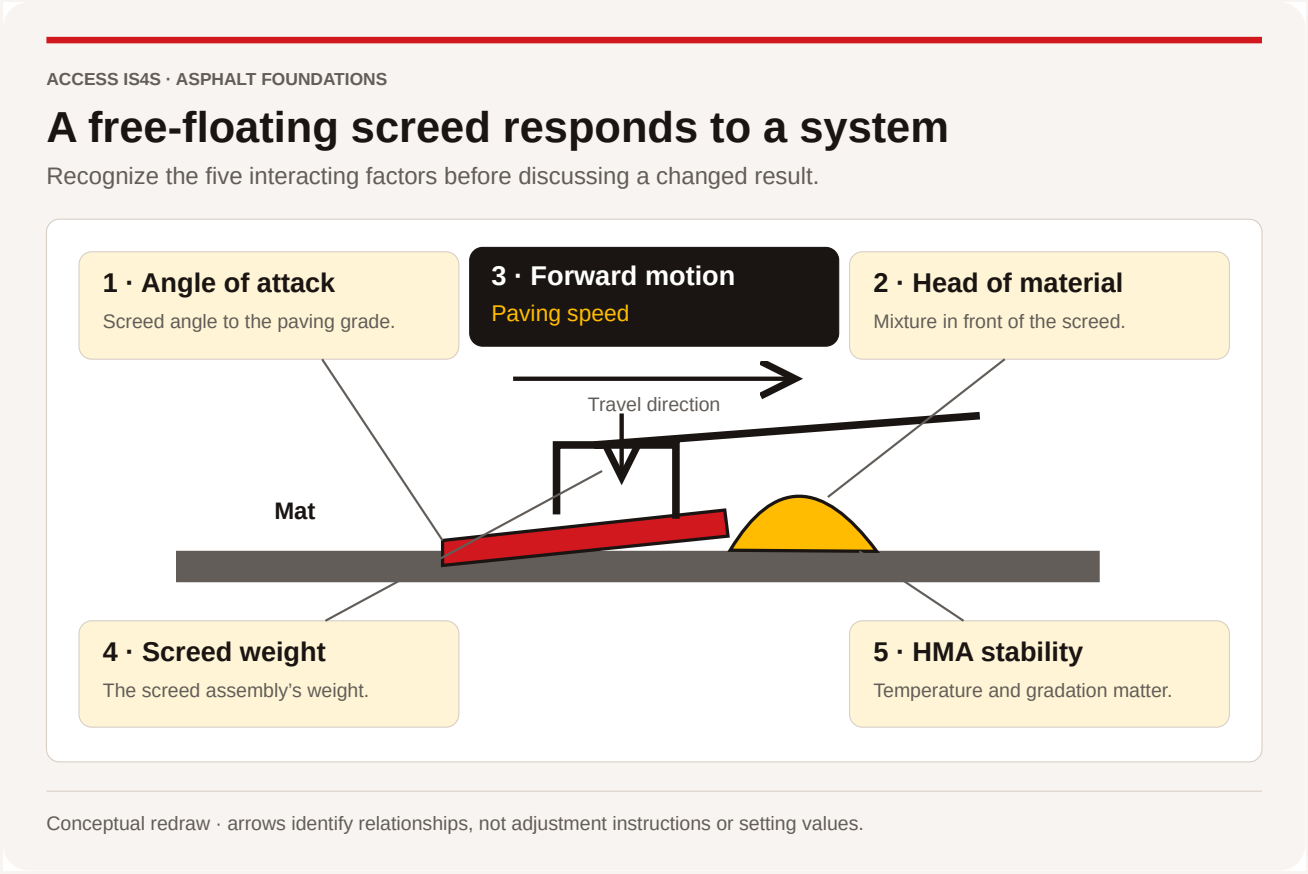


Figure 8.16 Free-Floating Screed

Note. From The asphalt handbook (7th ed., MS-4, Figure 8.16, p. 383; source PDF 417), by Asphalt Institute, 2007. Copyright 2007 by Asphalt Institute. Faithful crop from the supplied scan; no labels altered. Original figure contributor: Volvo Construction Equipment (credit on PDF 804). Source and context.

FIGURE T11

A free-floating screed responds to a system



Note. Adapted from Asphalt Institute. (2007). The asphalt handbook (7th ed., MS-4). Figure 8.16, p. 383 / PDF 417. Mirrored and simplified the schematic; grouped the five factors in numbered callouts; omitted adjustment procedures and numeric settings. Original contributor: Volvo Construction Equipment. Copyright 2007 by Asphalt Institute. ACCESS redraw (2026). Source and context.

Explain a relationship without guessing a setting

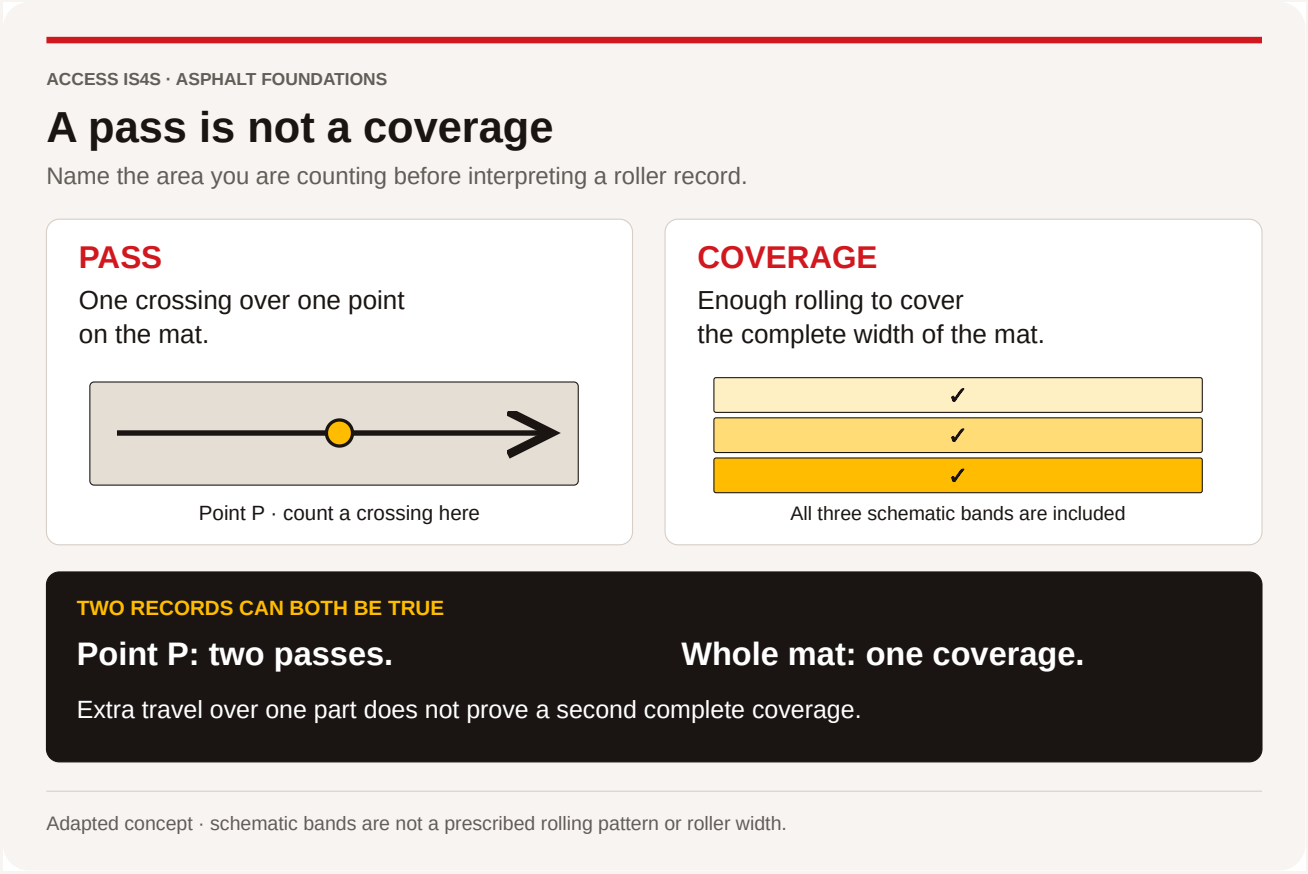
Match five factors in the source picture to the redraw. A fictional learner says, “The screed is heavy, so the layer must always be the same thickness.” Explain why the diagram does not support that conclusion. Identify which machine information and supervised instruction would be needed before changing a control. **Skills:** Reading, Problem Solving, Communication.

Count an observation correctly

A **pass** counts a roller crossing over a point. **Coverage** concerns reaching the full intended width. These are different observations. A record that counts repeated travel over one point cannot establish that every part of the mat received the same work. Neither count replaces the required density evidence.

FIGURE T12

A pass is not a coverage



Note. Adapted from Asphalt Institute. (2007). The asphalt handbook (7th ed., MS-4). Figure 9.1, p. 416 / PDF 450. Replaced the original five-pass pattern with separate point and full-width concepts, plus a new record-reading example. No operating pattern is prescribed. Copyright 2007 by Asphalt Institute. ACCESS redraw (2026). Source and context.

Check the claim in a roller record

Fictional record: point P was crossed twice; the left strip was crossed twice, the centre strip once and the right strip not at all. Can the writer claim complete width coverage? Explain what the record supports and what remains unknown. Design a clearer record with location, time, observed action and a space for the required test reference. **Skills:** Numeracy, Problem Solving, Creativity and Innovation, Writing.

4 · Working in a paving team

SKILLEDTRADESBC · LINE A

A4 · Describe industry expectations of/for workers

Read the trade words, use them in a clear explanation, then keep evidence of your own work.

Outline learning task	Practise here
A4.1 · Explain the importance of Health and Safety	“Choose the responsible next step”
A4.2 · Define types of paving jobs	“Make a role and dependency map”
A4.3 · Explain license requirements for paving equipment operators	“Separate four kinds of permission”
A4.4 · Demonstrate professionalism	“Repair a rushed handover”
A4.5 · Explain employer’s responsibility to provide a safe environment	“Choose the responsible next step”
A4.6 · Describe advancement opportunities	“Prepare a useful enquiry and growth plan”
A4.7 · Describe job security	“Check time and revise the plan” through “Prepare a useful enquiry and growth plan”
A4.8 · Explain role of organized Labour	“Prepare a useful enquiry and growth plan”
A4.9 · Review working conditions	“Check time and revise the plan” through “Prepare a useful enquiry and growth plan”

Connect the job title to the work

Outline job term	Broad function to explain	Information another role needs
Raker	Supports hand placement and finishing	Where the agreed handwork is needed
Screed	Forms the placed mat as part of the paver system; the screed role attends to this work	The applicable placement requirements and changes
Asphalt Paver	Places the supplied mixture through the machine system	Delivery status and the approved work arrangement
Compaction Roller	Compacts the placed material under the assigned procedure	Placement status and the applicable compaction requirements
Milling machine	Removes or profiles existing pavement	The authorized removal or profile requirements

Worked language check. “The roller does the road” is hard to check. “The compaction roller compacts the placed material” names the equipment and its function. In “Make a role and dependency map”, make each explanation this specific, then add the information handover. These are broad functions, not machine operating instructions.

Understand the role and the handover

A paving team coordinates people whose tasks depend on one another. The driver delivers material; the paver operator manages the machine's placement function; the screed role attends to the screed's contribution to the placed mat; rakers support hand placement and finishing; roller operators perform the assigned compaction work; milling operators work with equipment that removes or profiles existing material. Job titles and boundaries vary. A job title alone does not establish a person's training, authorization or responsibility on a particular site.

Fictional team brief W1. At the Willow Reach training centre, a classroom team is preparing an explanation of a paving job. Jo coordinates the display. Ren checks the delivery/placement sequence. Ash prepares the screed-and-raker explanation. Kim checks the compaction and milling distinctions. The team has public introductory descriptions, but no site plan, machine manuals or permission to enter a worksite. The display must explain broad roles and show where one role needs information from another.

Make a role and dependency map

Map haul delivery, paver, screed, raker, roller and milling roles. For each, write one broad function and one information handover. Identify two assumptions W1 does not authorize. Each learner explains a different role before the group combines the map. **Skills:** Reading, Communication, Collaboration.

Treat health and safety as a shared system with defined duties

Health and safety includes planning, suitable equipment and systems, instruction, supervision, reporting and review. It is not satisfied by telling a worker to “be careful.” A new person needs to know the supervisor, relevant hazards, emergency arrangements and how the actual task is done safely. Employers have responsibilities that cannot be replaced by a worker's confidence or a classroom certificate.

B.C.'s orientation provisions require workplace-specific training before a young or new worker begins work. A new worker can be an experienced person who is new to that workplace or affected by changed hazards. Additional orientation and training must be provided when observation shows a need or when the young or new worker requests it. Records are required. The complete workplace program and applicable legislation must guide actual work.

Fictional case W2. A learner arrives for a first workplace visit after completing classroom vocabulary. A host says, “You know the equipment names, so join that crew.” No supervisor contact, task demonstration or emergency information has been provided. Another host offers to stop the assignment and arrange the workplace orientation. The learner wants to participate but does not know the site's communication arrangements.

Choose the responsible next step

Explain why the second host's approach addresses a real gap. Draft a brief request for orientation and task clarification. Identify information the employer must provide and a question the learner can ask. Do not improvise signals or a route through live equipment. **Skills:** Problem Solving, Writing, Communication.

Check licences and authorization precisely

A driver's licence, admission to a training course, equipment competence and site authorization answer different questions. ICBC identifies Class 5 as the common passenger-vehicle licence. A provider may publish its own admission requirement. Neither statement proves that the holder may operate every paving machine or drive every vehicle combination. Determine the actual vehicle, activity and location, then verify the applicable licence class, restrictions, endorsements, training and employer authorization through the responsible sources.

Document-check case W3, using a real provider statement and fictional decision. IUOE Local 115's public Asphalt Paving Laydown Technician page, retrieved 29 October 2026, lists a valid Class 5 or 7 driver's licence among its admission prerequisites. A fictional learner reads this and says, “Then Class 7 authorizes me to drive any haul truck at work.” The page is a course-admission source; it is not a legal ruling on that vehicle or a workplace assignment.

Separate four kinds of permission

Create a table for course admission, driver's licence, equipment competence and site assignment. State the question each answers and who or what record can confirm it. Correct the learner's claim without replacing it with another unsupported universal rule. **Skills:** Reading, Problem Solving, Writing.

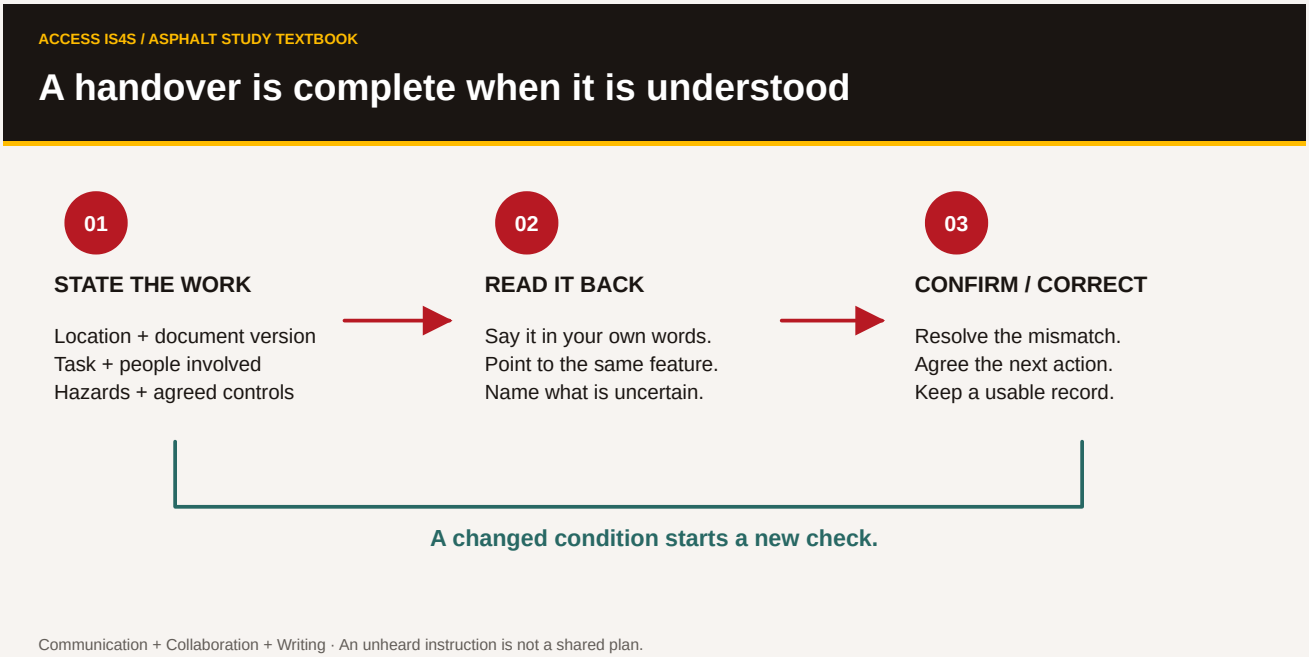
Practise professionalism through observable actions

Professionalism includes preparing for the agreed work, communicating changes, checking information, respecting people and taking care with records and materials. Pride in a project can be shown through a correction made before a mistake is passed on. It does not require pretending to know something or remaining silent when instructions are unclear.

Worked handover. “The map is finished” leaves the receiver guessing. A useful handover is: “The haul, paver and roller functions match our classroom brief. The licensing question is unresolved. The source note is saved as version 2. Please check the role labels; do not describe the draft as a site procedure.” The receiver restates the completed and pending parts before continuing.

FIGURE T13

A handover is complete when it is understood



Note. ACCESS IS4S. (2026). A handover is complete when it is understood [Original instructional SVG]. Conceptual illustration; read the stated conditions in the lesson.

Repair a rushed handover

Jo says, “Everything is approved—just print it,” although W3 is still unresolved. Give a respectful correction that protects the deadline and the accuracy of the work. Agree a revised print note and record each person's contribution. **Skills:** Communication, Collaboration, Adaptability.

Plan development without promising job security

A worker may develop from entry tasks toward equipment roles, specialist functions, supervision or other pathways through training, experience and appropriate assessment. Progress is not automatic or identical for everyone. Ask what evidence, supervision and opportunities the actual employer or provider requires. A forecast of industry demand cannot guarantee a particular person's employment. Season, project timing, skills, location, employer needs and economic conditions can affect work.

Paving work may involve outdoor conditions, dust and fumes, noise, night or weekend schedules and seasonal changes. A job enquiry should ask about the actual schedule, hazards, controls, travel, orientation and employment terms. Do not assume a fictional schedule establishes lawful hours, overtime or pay rules.

Fictional schedule W4. A classroom planning record lists a Monday period from 18:00 to 02:00, including a 30-minute unpaid break for arithmetic practice. Tuesday is cancelled because of a changed project plan. Wednesday has six paid hours. These are invented conditions, not a recommended work arrangement or statement of employment law.

Check time and revise the plan

Find Monday's elapsed and paid time, then the total paid hours recorded for Monday and Wednesday. Write two questions about a real employer's terms that this classroom record cannot answer. Revise a weekly learning plan so a cancelled day does not become falsely recorded practice. **Skills:** Numeracy, Adaptability, Writing.

Understand organized labour and employment documents

A union can represent workers in collective bargaining with an employer. A collective agreement records negotiated terms for the bargaining relationship; its application depends on the employer, bargaining unit, dates and the actual agreement. B.C.'s Labour Relations Board maintains a searchable directory of filed agreements. An old agreement found online is not automatically the current agreement for a new job. A training provider's fee page is also different from an employment agreement.

Prepare a useful enquiry and growth plan

Create an enquiry with four questions: applicable employment/collective-agreement information; expected working conditions; training and authorization for a desired role; and how progression is assessed. Add one sentence explaining collective bargaining and one explaining why industry demand or course attendance does not guarantee employment. Add a two-step learning goal with an observable result and a check date. Use fictional names; do not send the enquiry. On a device, save and reopen it with a clear version name. A partner suggests one improvement and you decide whether to use it. **Skills:** Digital, Writing, Creativity and Innovation, Communication.

5 · Read road documents and quantities

SKILLEDTRADESBC · LINE A

A5 · Describe basic road and highway profiles

Read the trade words, use them in a clear explanation, then keep evidence of your own work.

Outline learning task	Practise here
A5.1 · Review categories of projects	“Sort use, ownership and authority”
A5.2 · Describe types of owners	“Sort use, ownership and authority”
A5.3 · Describe common types of asphalt work	“Describe the road form”
A5.4 · Review specifications for different types of roads	“Sort use, ownership and authority”, “Recalculate after the revision”
A5.5 · Describe cross section of road	“Choose the right view” through “Describe the road form”

Identify the project before the specification

A project may be described by use—commercial, industrial or residential—or by jurisdiction and delivery, such as a provincial highway project. These categories answer different questions and can overlap. A private industrial site is different from a provincially administered highway, even if both use asphalt. Owners can include provincial, federal, municipal or private organizations. A funder, designer, contractor and owner may be different parties.

Fictional practice project R1 · Cedar Learning Access, drawing P-01 revision B. The fictional Cedar Learning Society owns a short access route on its own practice property. Valley Surface Ltd. is the fictional contractor. The classroom project brief classifies the work as a private facility access project. A fictional community fund contributes money but does not become the owner. There is no real construction instruction or permission attached to this case.

The practice document register contains P-01 revision B, a plan/profile/section sheet; Q-01 revision B, a quantity record; and SP-01 revision B, a teaching specification card. An earlier P-01 revision A is retained for comparison and marked superseded. In a real project, the contract and authorized document system determine applicable requirements; a familiar document title does not establish that it governs every road.

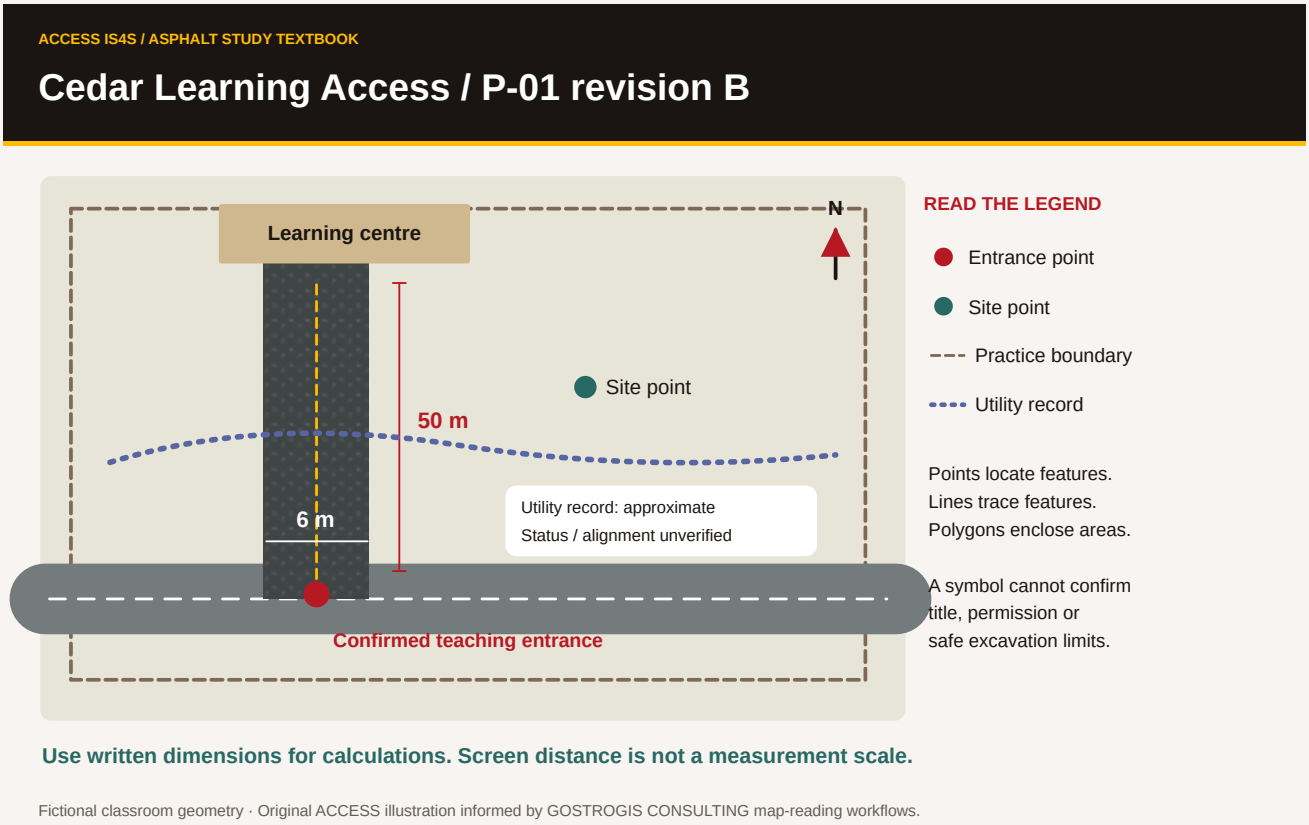
Document families include B.C.'s provincial highway specifications, **Master Municipal Construction Documents (MMCD)**, a local government's project requirements, and building-specific architectural or civil drawings and specifications. MMCD's publisher produces standardized municipal construction documents and updates. A municipal or private project may adopt particular documents through its contract; naming MMCD does not establish that every edition or clause applies. Provincial specifications likewise need the applicable contract version and special provisions. Record the issuer, edition, project adoption and any authorized change before relying on a requirement.

Sort use, ownership and authority

Identify the project use, owner, contractor and funder in R1. Explain why “private” and “asphalt” do not tell you which public standard or special provision applies. Compare provincial, MMCD, local-government and building-specific document families: what project-adoption evidence would you check before using any of them? Name the record you would need before treating a requirement as authorized. **Skills:** Reading, Problem Solving, Writing.

FIGURE T14

Cedar Learning Access / P-01 revision B



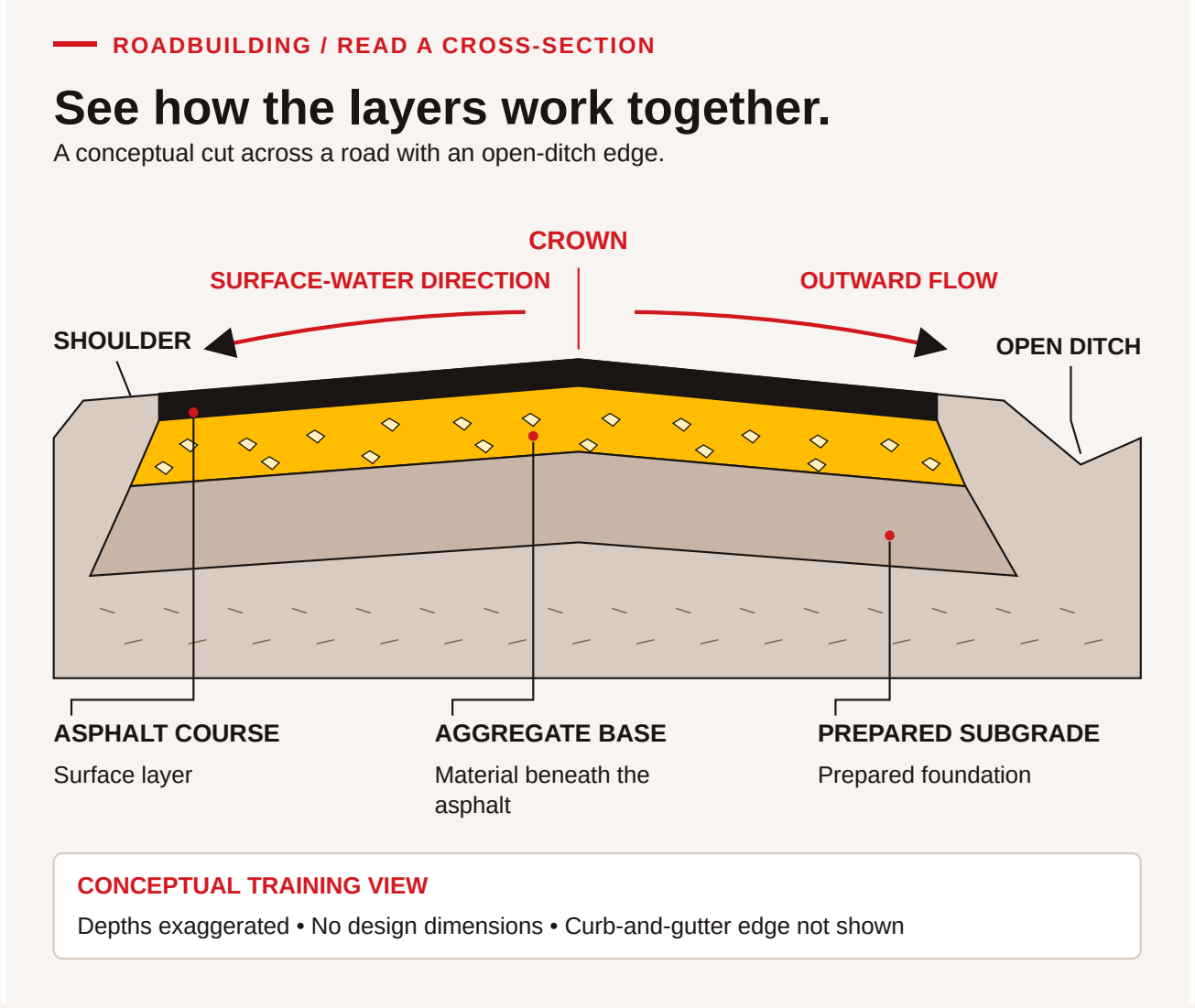
Note. ACCESS IS4S. (2026). Cedar Learning Access / P-01 revision B [Original instructional SVG]. Informed by GOSTROGIS. (2026). Site planner, parcel site planner, map symbology and utility-network documentation (commit 9e409ba). All geometry is fictional and not georeferenced.

Read three views of the same place

A **plan** looks down on a project and shows its horizontal arrangement. A **profile** shows elevations along a chosen line, commonly using chainage or stationing to locate points. A **cross-section** cuts across the route and shows its transverse form and layers. A number must be associated with the correct view, direction and unit. Plan width, profile rise and layer thickness are different quantities.

Figure LA-2

See how the layers work together

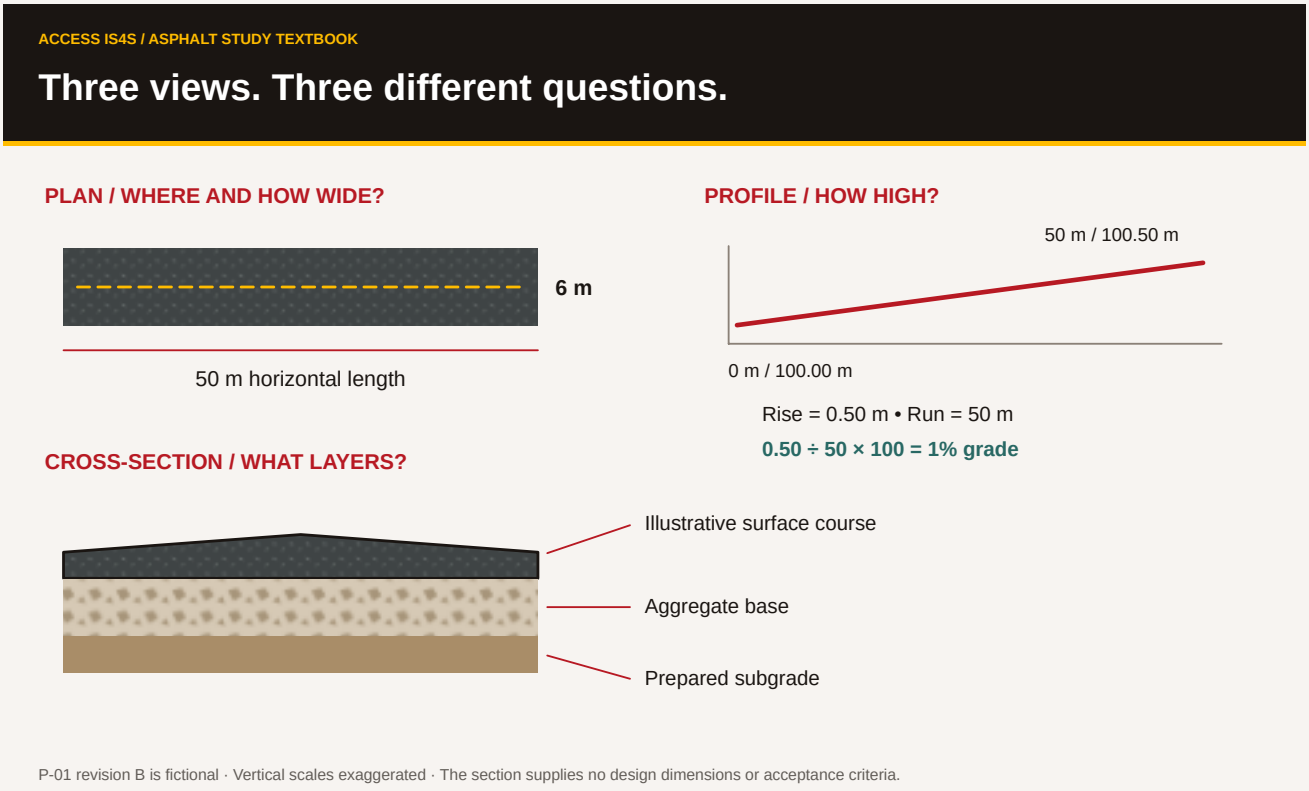


Note. An original ACCESS IS4S instructional illustration. Layer depths are exaggerated. Curb-and-gutter and superelevation alternatives are discussed in the text, not drawn here. No design dimensions or acceptance standards are supplied. By ACCESS IS4S (2026b); see the figure references in the working reference section.

A **crown** is a raised transverse form that can direct water toward the sides. **Superelevation** describes a cross-slope arrangement used on curves; its design depends on the project. **Runoff** is water moving from the surface. An open ditch and shoulder arrangement differs from a contained curb-and-gutter edge. A sidewalk serves a different function from the vehicle surface. A grader is equipment used in earthwork or grading, not a layer in the finished section. Drainage and preparation must be coordinated before paving; a drawing needs more than a surface outline to establish a buildable design.

FIGURE T15

Three views. Three different questions.



Note. ACCESS IS4S. (2026). Three views. Three different questions. [Original instructional SVG]. Conceptual illustration; read the stated conditions in the lesson.

Distinguish the supporting layers

A **base** and **subbase** are supporting layers where the design includes them. The **prepared subgrade** is the prepared ground beneath the pavement structure. Figure LA-6 separates one unbound-base arrangement so you can name the layers. Other structures exist, including full-depth asphalt; a subbase is not automatically present on every road. This teaching illustration supplies no dimensions for P-01.

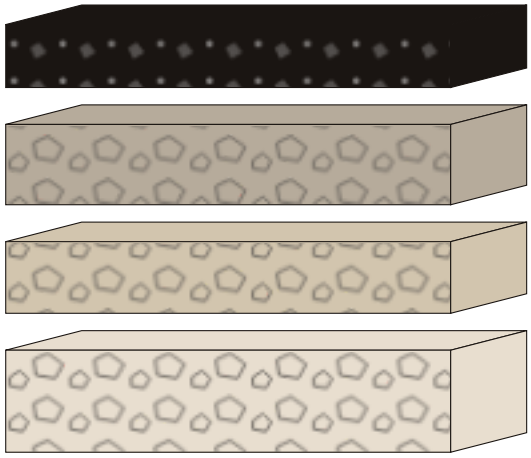
Figure LA-6

Name the layers

ACCESS IS4S · ASPHALT FOUNDATIONS

Name the layers

One example: asphalt pavement on an unbound base, with a subbase.



1. **Asphalt surface**
   The upper asphalt layer.
2. **Granular base**
   Supports the asphalt surface.
3. **Subbase, where specified**
   An additional supporting layer.
4. **Prepared subgrade**
   The prepared ground beneath the structure.

Conceptual section · other pavement structures exist · layer thicknesses are not a design.

Note. Adapted from *The asphalt handbook* (7th ed., MS-4; Figure 1.6, p. 14 / PDF 48), by Asphalt Institute, 2007. Selected the unbound-base example, separated its layers and added functional descriptions. Other structures and design dimensions are not implied. Copyright 2007 by Asphalt Institute; original contributor credits retained.

FIGURE T16

Compare the handbook's pavement structures

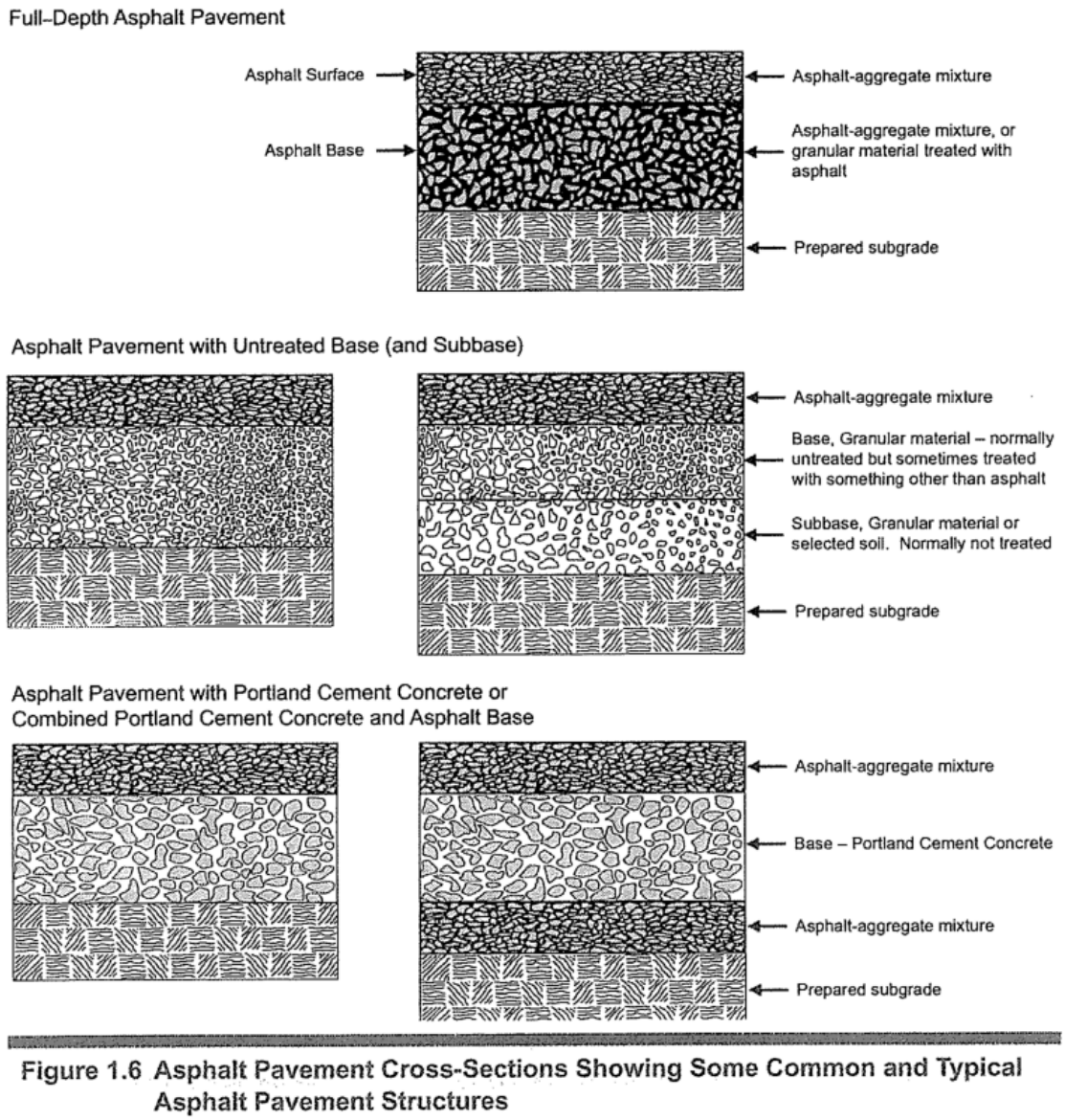


Figure 1.6 Asphalt Pavement Cross-Sections Showing Some Common and Typical Asphalt Pavement Structures

Note. From The asphalt handbook (7th ed., MS-4, Figure 1.6, p. 14; source PDF 48), by Asphalt Institute, 2007. Copyright 2007 by Asphalt Institute. Faithful crop from the supplied scan; no labels altered. Source and context.

P-01 revision B · Complete fictional drawing data. The rectangle extends from chainage 0 m to 50 m. Its constant width is 6 m. The new uniform layer is 40 mm thick for this calculation exercise. No widening, taper, overlap, waste allowance or material loss is included. The exercise asks for geometric volume only. Use the written dimensions; a screen image is not a measuring scale.

Profile point	Chainage along centreline	Illustrative elevation
Start	0 m	100.00 m
Midpoint	25 m	100.25 m
End	50 m	100.50 m

The section names surface course, aggregate base and prepared subgrade. Their presence in a conceptual sketch does not specify material properties, layer thicknesses other than the given calculation layer, drainage design or acceptance criteria. All elevations and dimensions here are invented. The profile arithmetic is not a recommendation for road grade or crossfall.

Choose the right view

For route length, width, centreline rise and layer thickness, identify the relevant view and unit. Explain why the profile elevation 100.50 m is not the layer thickness. Sketch three boxes labelled plan, profile and cross-section and put the matching information in each. **Skills:** Reading, Numeracy, Creativity and Innovation.

Describe the road form

Using the visual and paragraph, explain crown, shoulder, open ditch, curb/gutter, sidewalk, runoff and superelevation. Put “grader” in the correct category. Use Figure LA-6 to distinguish granular base, subbase where specified and prepared subgrade; explain why one illustrated arrangement does not establish the layers on every project. Choose one drainage question and one preparation question that the conceptual drawing leaves unanswered. **Skills:** Reading, Communication, Problem Solving.

Point, name and explain

Use the drawing and the reading to connect each word to a function. A learner may point, annotate a copy, or describe the location verbally.

Trade word	Look for or explain	What the word does not tell you
Crown	Raised transverse form near the centre of a crowned road	The specified crossfall or an acceptance result
Shoulder / open ditch	Edge area / open drainage channel	Their required dimensions or capacity
Curb and gutter	A contained edge and associated drainage form	That it is drawn in this open-ditch example
Runoff	Surface water moving away from where it falls	The complete route, quantity or drainage design
Superelevation	A curve-related cross-slope arrangement	That it is the same shape as the crown in this figure
Grader	Equipment used in grading work	A layer in the cross-section
Prepared subgrade	The prepared foundation below the pavement structure	That its preparation has been inspected or accepted

Worked language check. “The arrows are runoff direction; the dark layer is the asphalt course.” The arrows describe movement; the layer describes material. In “Describe the road form”, explain two more features in the same way.

Calculate a quantity you can check

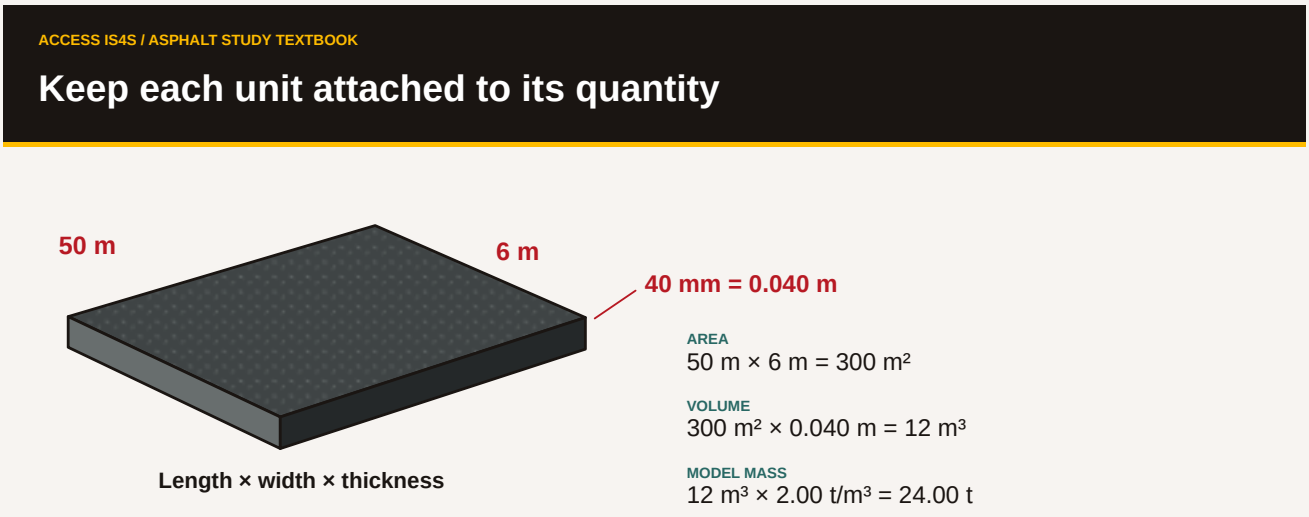
Worked example, P-01 revision B. Convert thickness first: `40 mm ÷ 1,000 = 0.040 m`. Plan area is `50 m × 6 m = 300 m²`. Geometric layer volume is `300 m² × 0.040 m = 12 m³`. Estimate: a 300 m² layer one-tenth of a metre thick would occupy 30 m³; a 0.04 m layer is less than half as thick, so 12 m³ is plausible. Units check as `m × m × m = m³`.

The profile rises `100.50 − 100.00 = 0.50 m` across 50 m of stated horizontal distance. For this drawing-reading exercise, rise/run is `0.50 ÷ 50 × 100 = 1%`. This centreline grade is not the transverse crown or an acceptance limit. A real quantity may need a different geometry and an approved measurement method; our rectangular model has stated simplifications.

Q-01 revision B · Fictional mass model and delivery records. For this exercise only, the instructor supplies a conversion factor of **2.00 tonnes per cubic metre**. It is an invented teaching factor, not an asphalt density specification. The planned model mass is `12 m³ × 2.00 t/m³ = 24.00 t`. Three classroom records list 8.00 t, 8.00 t and 7.60 t. These are fictional record inputs, not a real delivery or placement measurement.

FIGURE T17

Keep each unit attached to its quantity



Note. ACCESS IS4S. (2026). Keep each unit attached to its quantity [Original instructional SVG]. Conceptual illustration; read the stated conditions in the lesson.

Check the quantity and record

Independently reproduce the area, thickness conversion, volume and model mass with units. Sum the three records and calculate their difference from the planned model mass. Explain why that numerical difference alone cannot establish placed thickness, compaction, payment entitlement or rejected work. **Skills:** Numeracy, Reading, Writing.

Use the applicable version and conditions

SP-01 revision B · Teaching specification card, fictional. Use the dimensions printed in the current classroom drawing. Record the revision with every calculation. Keep calculation quantities distinct from acceptance measurements. Refer conflicting instructions to the instructor acting as document coordinator before issuing a final classroom quantity note. This is the complete specification for the exercise; it does not adopt a real road specification.

Change notice R2, fictional. The coordinator issues P-01 revision C, extending the rectangle to 60 m while retaining 6 m width and 40 mm layer thickness. The endpoint illustrative elevation becomes 100.60 m; the same linear 1% profile is used for arithmetic practice. The mass model remains 2.00 t/m³. The three existing classroom records stay unchanged. Revision C replaces B for the new quantity calculation; retain B as the previous version.

Recalculate after the revision

Find the new area, volume and model mass, and the increase from revision B. Reconcile the unchanged record total with the new model requirement. Explain why changing a filename without recalculating would leave misleading information. Make a versioned paper or digital quantity note with inputs, method, check, result and unresolved limits. **Skills:** Numeracy, Adaptability, Writing; Digital when performed.

Review and hand over

Work in pairs: one checks geometry and units, the other checks document revision and claims. Swap responsibilities on the revised case. Give a handover that states the current revision, quantity, supplied factor, record total and the decision that remains outside this exercise. Each learner keeps a checked calculation and records one useful correction or confirmed detail. **Skills:** Collaboration, Communication, Problem Solving.

Read the map record behind the symbol

A **point** can identify an entrance or a mapped site location. A **line** can represent a road or a utility record. A **polygon** encloses an area, such as a parcel or project selection. The same coloured dot can be used for very different features; read its legend and attributes. Changing a symbol's colour does not change its position or legal meaning.

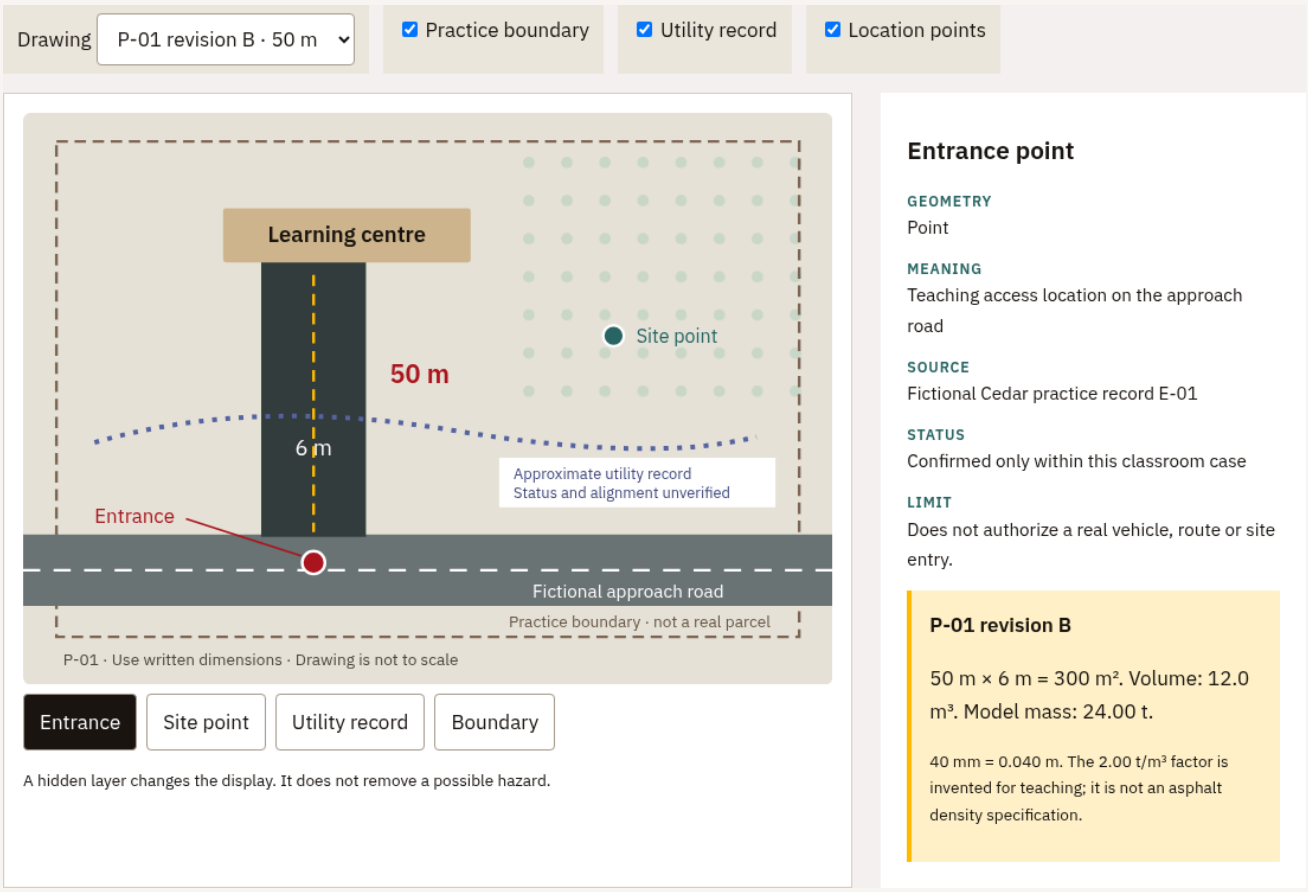
An **entrance** is an access location. A **site point** may locate a facility or represent a record. A **parcel boundary** describes a parcel geometry from a particular source. None of those is interchangeable. A point in the middle of an area does not tell a delivery driver where to enter. A project-selection area filters what is displayed; it does not create or alter a legal boundary.

A useful map-reading record keeps the feature type, source, date, status and uncertainty together. A missing value is not zero. “Unknown utility status” does not mean “no utility.” A source edit date does not prove when a buried line was installed, surveyed or last verified. Separate ownership, operation and maintenance responsibility when the record provides different fields.

Open the Cedar plan desk. It contains fictional features for this lesson. Toggle a layer, select a record, change the drawing revision and export a note. The desk's written dimensions drive the quantity model; no geographic coordinates or real land rights are attached to it.

FIGURE T18

Inspect the record behind the map symbol



Note. ACCESS IS4S. (2026). Cedar Learning Access plan desk [Actual interface screenshot]. Fictional classroom geometry. Map-reading concepts adapted from GOSTROGIS CONSULTING documentation, commit 9e409ba, with owner authorization. Source and context.

Worked map decision. A proposed driver instruction says, “Go to the site dot.” The map shows a site point inside the practice boundary and a separate entrance point on the road. A clearer instruction identifies the confirmed entrance and the current access arrangement. If routing information is used, it must match the vehicle and task: a public car route does not establish truck access, loading restrictions or permission. Road travel distance differs from a straight-line distance between symbols.

Read, toggle and preserve a record

In the plan desk, select the site point, entrance and utility record in turn. For each, record what it means, its stated status and one thing it cannot establish. Hide and show the utility layer: explain why hiding a feature does not remove the real-world hazard it could represent. Export your own note, reopen the file and check that it retains the selected revision and record. **Skills:** Reading, Digital, Writing, Problem Solving.